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ED3-EE GROUP 24

PROJECT TOPIC: AUTOMATED SECURITY CHECKPOINT SYSTEM

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TABLE OF CONTENTS

Table of Contents ……

PROJECT OVERVIEW **3**

IMPORTANCE OF THE PROJECT **3**

- Enhanced Security 3
- Improved Traffic Flow 3
- Reduced Human Error 3
- Time Efficiency 3
- Operational Cost Reduction 3

METHODOLOGY AND DESIGN APPROACH **4**

Step 1: Analysis of Manual Security Checkpoint Systems 4

Step 2: Development of Functional Block Diagram 4

BLOCK DIAGRAM 4

Step 3: Selection of Components 4

Step 4: Design of the circuit diagram 5

Step 5: Code Implementation and System Operation 5

C++ CODE UPLOADED INTO ARDUINO UNO 6

Step 6: Testing and Validation 8

SIMULATION LINK 8

PROJECT ARCHITECTURE **9**

Overall, the architecture ensures a smooth flow of operation: Input (RFID) → Processing (Arduino Uno) → Output (Servo motor, LEDs, Buzzer), forming a reliable and efficient automated checkpoint system **9**

CONCLUSION **10**

REFERENCES **10**

PROJECT OVERVIEW

Many existing security checkpoints rely heavily on manual methods to inspect, identify, and verify vehicles. While this approach has been widely used, it is often slow, prone to human error, and can result in unauthorized vehicles gaining access. Additionally, manual inspection contributes to traffic congestion, especially in high-demand areas such as campuses, residential communities, and restricted facilities.

This project focuses on the design and implementation of an automated checkpoint system that enhances both security and efficiency using Radio Frequency Identification (RFID) technology. The system is intended to automatically identify vehicles equipped with authorized RFID tags as they approach the checkpoint. Once detected, the system verifies the vehicle's credentials against a stored database in real time.

Authorized vehicles are granted access seamlessly, allowing for smooth and uninterrupted movement through the checkpoint. In contrast, vehicles without valid authorization are denied entry, and appropriate alerts or actions can be triggered. This automation significantly reduces the need for human intervention, minimizes delays, and improves overall reliability.

By integrating RFID technology with control mechanisms such as automated barriers and microcontroller-based processing units, the system aims to provide a secure, fast, and efficient alternative to traditional checkpoint operations. Ultimately, this project demonstrates how modern technology can be leveraged to improve security infrastructure while optimizing traffic flow.

IMPORTANCE OF THE PROJECT

- **Enhanced Security**
  The system ensures that only vehicles with valid RFID authorization are granted access, significantly reducing the risk of unauthorized entry and improving overall security at checkpoints.
- **Improved Traffic Flow**
  Automation minimizes delays caused by manual inspection, allowing vehicles to pass through checkpoints quickly and efficiently, thereby reducing congestion.
- **Reduced Human Error**
  By replacing manual verification with an automated process, the system eliminates errors associated with fatigue, oversight, or misjudgment by security personnel.
- **Time Efficiency**
  The RFID-based system performs identification and verification in real time, saving time for both operators and vehicle users compared to traditional methods.
- **Operational Cost Reduction**

  Over time, the system reduces the need for a large number of security personnel, leading to lower labor costs and more efficient resource utilization.

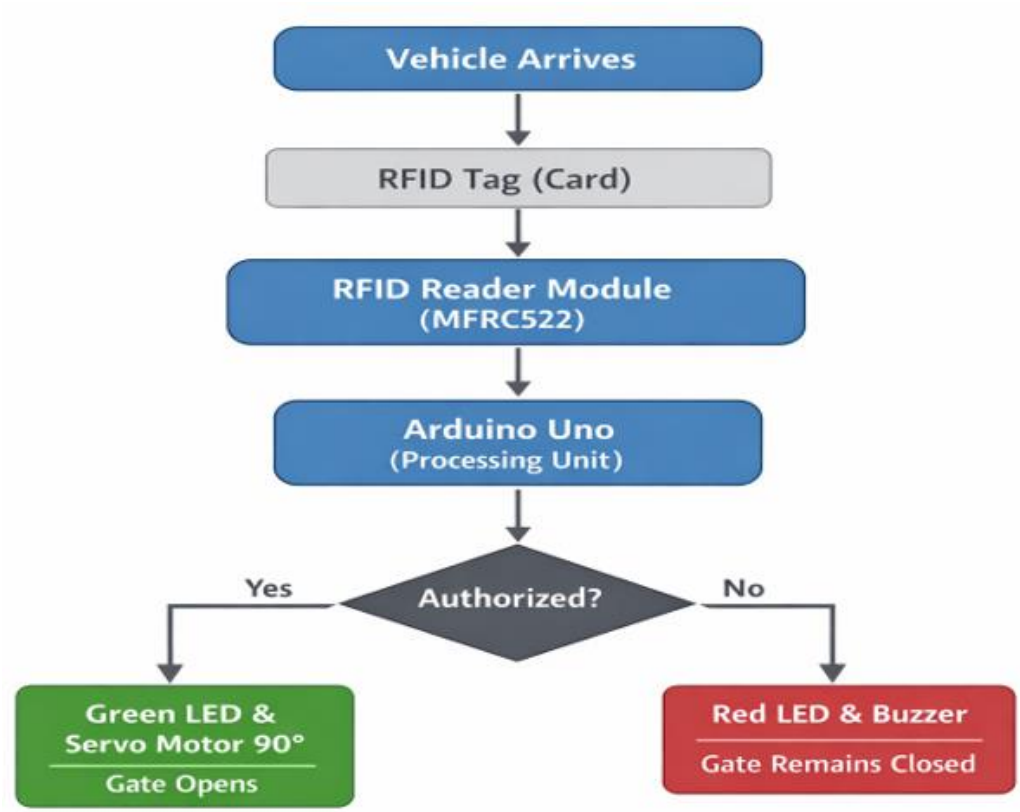
METHODOLOGY AND DESIGN APPROACH

The design and development of the automated checkpoint system followed a structured methodology to ensure efficiency, reliability, and proper functionality. The approach consisted of the following stages:

Step 1: Analysis of Manual Security Checkpoint Systems
An initial study was conducted on existing manual checkpoint systems to identify their limitations, such as delays, human error, and the possibility of unauthorized access. This analysis provided a clear understanding of the challenges and served as a foundation for designing a more efficient automated system.

Step 2: Development of Functional Block Diagram
A functional block diagram of the automated checkpoint system was designed to represent the overall system architecture. This diagram illustrated the sequence of operations, starting from vehicle detection, RFID scanning, data verification, and finally access control. It helped in visualizing how each component interacts within the system.

BLOCK DIAGRAM



Step 3: Selection of Components
Based on the functional requirements, appropriate hardware components were selected for both simulation and implementation. These include:

- Arduino Uno (microcontroller)

- MFRC522 RFID module
- RFID cards and key fobs
- Breadboard
- Servo motor (for barrier control)
- Green and Red LEDs
- Buzzer
- 220Ω resistors
- Jumper wires

Each component plays a specific role in ensuring proper system operation, from identification to access control and user feedback.

Step 4: Design of the circuit diagram

After obtaining the necessary components, the functional block diagram provided guidance as to how the various circuits elements should be connected. The schematic diagram is shown in figure1.

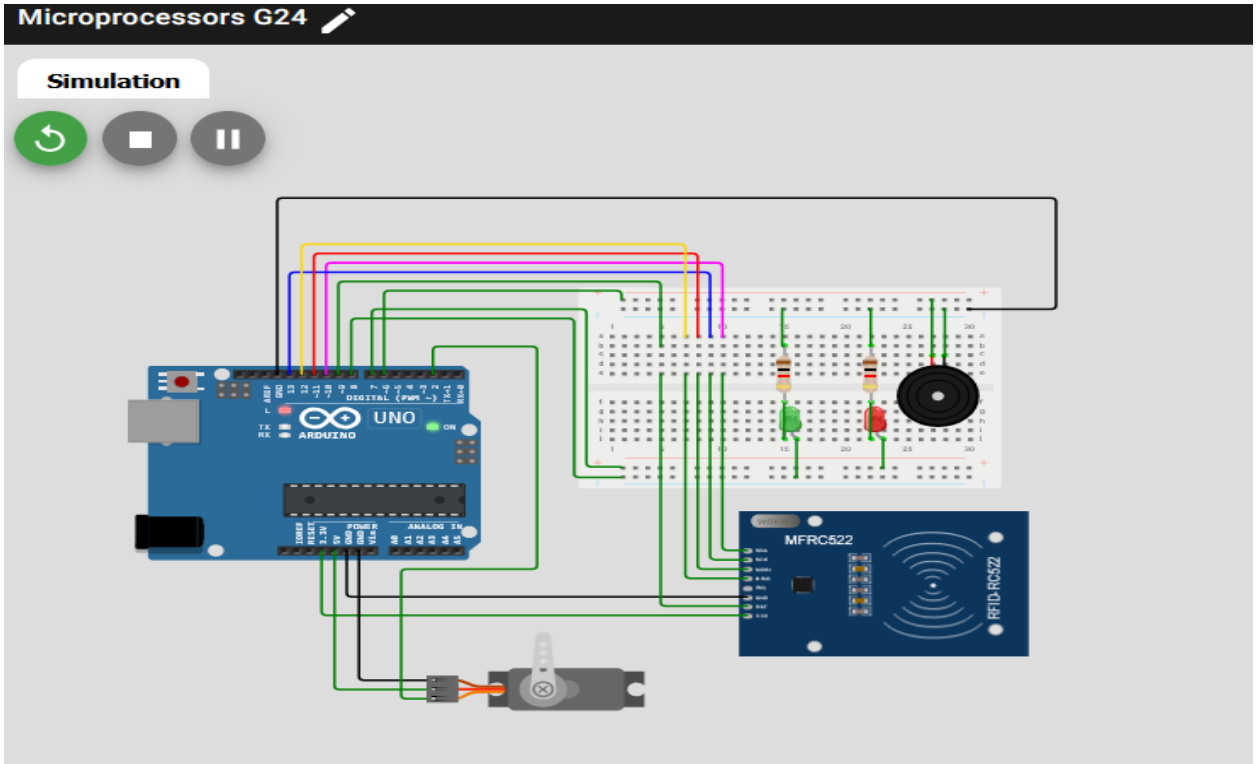


Figure1: schematic diagram of the automated checkpoint

Step 5: Code Implementation and System Operation

A C++ program was developed and uploaded to the Arduino Uno to control the system based on the defined specifications. The system operates as follows:

- When a vehicle approaches the checkpoint, the RFID reader scans the RFID tag.
- If the scanned UID matches one of the authorized IDs stored in the system:
    - Access is granted

    - The green LED turns on
    - The servo motor rotates 90° to open the barrier gate
- If the UID is not recognized:
    - Access is denied
    - The red LED turns on
    - The buzzer sounds
    - The barrier gate remains closed

The system code (as provided) was compiled and uploaded to the microcontroller to achieve this functionality.

C++ CODE UPLOADED INTO ARDUINO UNO

```
#include <SPI.h> // SPI communication library

#include <MFRC522.h> // RFID library

#include <Servo.h> // Servo library

#define SS_PIN 10 // RFID SS pin

#define RST_PIN 9 // RFID reset

#define LED_PIN1 8 // Access granted LED(Green)

#define LED_PIN2 7 // Access denied LED(Red)

#define SERVO_PIN 2 // Servo pin

#define BUZZER_PIN 6 // Buzzer pin

MFRC522 mfrc522(SS_PIN, RST_PIN); // RFID object

Servo gateServo; // Servo object

void setup(){

  Serial.begin(9600); // Start serial monitor

  SPI.begin(); // Start SPI

  mfrc522.PCD_Init(); // Initialize RFID reader

  pinMode(LED_PIN1, OUTPUT); // LED1(Green) output

  pinMode(LED_PIN2, OUTPUT); // LED2(Red) output

  pinMode(BUZZER_PIN, OUTPUT); // Buzzer output

  gateServo.attach(SERVO_PIN); // Attach servo

  gateServo.write(0); // Gate closed

  Serial.println("Scan your RFID card..."); // User prompt

}

void loop(){

  if(!mfrc522.PICC_IsNewCardPresent()) return; // Check card
```

```
if(!mfrc522.PICC_ReadCardSerial()) return; // Read card UID
Serial.print("UID tag: "); // Print UID label
String content = ""; // Store UID
for(byte i = 0; i < mfrc522.uid.size; i++){ // Loop through UID bytes
  if(mfrc522.uid.uidByte[i] < 0x10){
    content.concat("0");   // Leading zero
  }
  content.concat(String(mfrc522.uid.uidByte[i], HEX)); // Convert to HEX
}
Serial.println(content); // Show UID
// List of valid card UIDs
String validCards[]= {"11223344", "01020304", "aabbccdd"}; // Authorized cards
int numberOfCards = 3; // Number of cards
bool accessGranted = false; // Access flag
for(int i = 0; i < numberOfCards; i++){ // Check cards
  if(content == validCards[i]){ // Compare UID
    accessGranted = true; // Valid card
    break; // Exit loop
  }
}
if(accessGranted){ // If card is valid
  Serial.println("Access Granted"); // Message
  digitalWrite(LED_PIN1, HIGH); // Turn on LED1(Green)
  gateServo.write(90);   // Unlock by rotating 90 degrees
  delay(3000); // Wait 3s
  gateServo.write(0);   // Lock again
  digitalWrite(LED_PIN1, LOW); // Turn LED(Green) off
}
else{ // If card not valid
  Serial.println("Access Denied"); // Message
  digitalWrite(LED_PIN2, HIGH); // Turn on LED2(Red)
  digitalWrite(BUZZER_PIN, HIGH); // Sound buzzer
```

```
    delay(3000); // Wait 3s
    digitalWrite(LED_PIN2, LOW); // Turn LED off
    digitalWrite(BUZZER_PIN, LOW); // Stop buzzer
  }
}
```

Step 6: Testing and Validation

Both the simulation and the physical implementation of the system were tested multiple times to ensure that the automated security checkpoint met the initial design specifications. Repeated testing helped verify system reliability, accuracy in authorization, and proper response of all components under different scenarios.

SIMULATION LINK

https://wokwi.com/projects/459798019162231809

PROJECT ARCHITECTURE

The architecture of the Automated Checkpoint System is designed based on the interaction of key hardware components, each performing a specific function to achieve seamless operation. The system follows a centralized control approach, where the microcontroller serves as the core processing unit, coordinating all inputs and outputs.

At the input stage:

- the **RFID system**—comprising the MFRC522 reader and RFID tags (cards/key fobs)—is responsible for vehicle identification. When a vehicle approaches the checkpoint, the RFID reader scans the tag and retrieves its unique identification (UID). This UID serves as the primary data used for authentication.
- The **Arduino Uno microcontroller** acts as the brain of the system. It receives the UID from the RFID reader via SPI communication and processes it by comparing it with a predefined list of authorized IDs stored in its memory. Based on this verification, the Arduino determines whether access should be granted or denied.

The output stage consists of multiple components that provide both physical control and user feedback:

- The **servo motor** functions as the barrier control mechanism. When access is granted, the Arduino sends a signal to rotate the servo motor by 90°, opening the gate. After a short delay, it returns to its original position to close the gate.
- The **Green LED** indicates successful authorization and access granted.
- The **Red LED** signals access denial.
- The **220 Ω resistors** are connected in series with the LEDs to limit the current flowing through them. This is important because LEDs are sensitive components that can be damaged by excessive current. The resistors ensure that only a safe amount of current reaches the LEDs, thereby preventing them from burning out and ensuring their longevity and reliable operation within the system.
- The **buzzer** provides an audible alert when an unauthorized vehicle attempts to gain access.

All components are interconnected using jumper wires on a breadboard, with resistors used to regulate current flow to the LEDs. The system operates in real-time, ensuring immediate response once a vehicle is detected and scanned.

Overall, the architecture ensures a smooth flow of operation: Input (RFID) → Processing (Arduino Uno) → Output (Servo motor, LEDs, Buzzer), forming a reliable and efficient automated checkpoint system

CONCLUSION

The design and implementation of the automated checkpoint system successfully demonstrate how modern technology can be used to improve security and operational efficiency. By replacing manual inspection methods with an RFID-based system, the project addresses key challenges such as delays, human error, and unauthorized access.

The system was able to accurately identify and verify vehicles using RFID tags, granting access only to authorized users while denying entry to unauthorized ones. The integration of the Arduino Uno, RFID module, servo motor, LEDs, and buzzer ensured smooth operation, clear user feedback, and effective control of the checkpoint barrier. Additionally, repeated testing confirmed that the system met the intended design specifications and performed reliably under different conditions.

Overall, this project highlights the effectiveness of automation in security applications. It provides a scalable and cost-effective solution that can be applied in various environments such as campuses, residential areas, and restricted facilities to enhance both safety and traffic flow.

Challenges encountered:

- Ensuring correct connections of the various components
- Synchronizing the servo motor, the two LEDS and buzzers during the operation of the system.

Possible improvements:

- Creating a database to store to more authorized tags or cards
- Including additional indicators such as a display to provide feedback to the drivers

REFERENCES

https://youtu.be/s6oQzemYMgg?si=q8N70rVBYuK1_08d